Advertising Linear Regression Analysis

📊 Project Title

Predicting Sales from Multi-Channel Advertising Spend
Linear Regression Analysis using Scikit-Learn

👨‍🎓 About me

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🎯 Project Objective

Develop and interpret a multiple linear regression model to predict product sales based on advertising expenditures across TV, Radio, and Newspaper channels. This analysis demonstrates core data science skills including data pre-processing, model training, coefficient interpretation, and model evaluation.

📋 Table of Contents

- Dataset Overview
- Methodology
- Model Results
- Key Insights
- Technical Implementation
- Model Evaluation
- Future Improvements
- Usage Instructions
- Requirements
- Academic Context

📦 Dataset:-

/kaggle/input/advertising-dataset/Advertising.csv'

Advertising dataset

Attribute	Description	Unit	Range
TV	TV advertising budget	$1000s	0.7 - 296.4
Radio	Radio advertising budget	$1000s	0.0 - 49.6
Newspaper	Newspaper advertising budget	$1000s	0.3 - 114.0
Sales	**Product sales**	**$1000s**	**1.6 - 27.0**

- **Size**: 200 observations
- **Source**: Classic Advertising dataset (UCI/Auto-generated)
- **File**: `advertising.csv`

🔬 Methodology :-

Multiple Linear Regression

Model Specification

```text
Sales = β0 + β1(TV) + β2(Radio) + β3(Newspaper) + ε
```

Where:

- β_0 = Intercept (baseline sales)
- β_1, β_2, β_3 = Coefficients for each advertising channel
- ε = Error term

Implementation Steps

1. **Data Loading** → pandas Data Frame
2. **Exploratory Analysis** → correlations, distributions
3. **Feature Preparation** → X matrix, y target
4. **Model Fitting** → `Linear Regression ().fit(X, y)`
5. **Coefficient Extraction** → `intercept_`, `coef_`
6. **Prediction & Residuals** → model diagnostics

📈 Model Results :-

```text
Model: Sales = [intercept] + [TV_coef]*TV + [Radio_coef]*Radio + [Newspaper_coef]*Newspaper

INTERCEPT: []
COEFFICIENTS:
├── TV: [] (Strongest predictor)
├── Radio: [] (Significant impact)
└── Newspaper: [] (Minimal/negative)
```

💡 Key Insights <a name="insights"></a>

1. **Radio advertising yields highest ROI per $1000 spent**
2. **TV provides consistent baseline sales lift**
3. **Newspaper shows negligible or negative sales impact**
4. **Combined channel strategy recommended over single-channel**

💻 Technical Implementation <a name="implementation"></a>

Core Code

```python
# 1. Setup
import pandas as pd
import numpy as np
from sklearn.linear_model import LinearRegression
import matplotlib.pyplot as plt
import seaborn as sns

# 2. Load and prepare data
df = pd.read_csv('advertising.csv')
X = df[['TV', 'Radio', 'Newspaper']]
y = df['Sales']

# 3. Train model
lm_model = LinearRegression()
lm_model.fit(X, y)

# 4. Extract results
print(f"Intercept: {lm_model.intercept_:.2f}")
print(f"Coefficients: {dict(zip(X.columns, lm_model.coef_))}")
```

Jupyter Workflow

```text
linear_regression.ipynb
├── 01_Data_Exploration.ipynb
├── 02_Model_Training.ipynb
├── 03_Results_Visualization.ipynb
└── README.md (this file)
```

📊 Model Evaluation

Metric	Value	Interpretation
R^2 Score		Explained variance ratio
Mean Absolute Error		Average prediction error ($1000s)
Residual Std		Model prediction consistency

```Python
from sklearn.metrics import r2_score, mean_absolute_error
y_pred = lm_model.predict(X)
print(f"R²: {r2_score(y, y_pred):.3f}")
print(f"MAE: {mean_absolute_error(y, y_pred):.3f}")
```

🚀 Usage Instructions

🔧 TOOLS AND LIB

```text
pandas>=2.0.0
NumPy>=1.24.0
scikit-learn>=1.3.0
matplotlib>=3.7.0
seaborn>=0.12.0
Jupiter>=1.0.0
```

🎓 Academic Context

Skills Demonstrated:

- Multiple linear regression theory and implementation
- Scikit-learn model fitting and evaluation
- Statistical coefficient interpretation

- Jupyter notebook best practices
- Technical documentation

🔮 Future ImprovementsOR REGULATIZATION

- ☐ Polynomial feature interactions (TV Radio)
- ☐ Ridge/Lasso regularization
- ☐ Cross-validation (K-Fold)
- ☐ Residual diagnostics plots
- ☐ Business ROI calculations
- ☐ Model deployment (Streamlet/Flask)

📞 Contact

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